Answers to Study Questions

CHAPTER 1

1. Sanitation means creating and maintaining hygienic and healthful conditions.
2. Substantive, which means they have the power of law and must be carried out.
3. The Food and Drug Administration (FDA). Meat and poultry products are not monitored by this agency.
4. Less expensive; better communication between regulators and firms.
5. Conflicting interests of other departments; monitor sanitation across the organization as a whole.
6. Meets regulatory requirements; protects the reputation of the brand and products; ensures that products are safe, of high quality, and free from contamination.

CHAPTER 2

1. Molds, yeasts, bacteria, and viruses. (See chapter for examples.)
2. Yeasts look slimy and creamy white; molds are fuzzy and cottonlike.
3. Lag phase—adapting to the environment.

   Logarithmic growth phase—cells multiply rapidly.

   Stationary growth phase—various factors limit growth.

   Accelerated death phase—lack of nutrients and buildup of waste products cause rapid death of microbial cells.

   Reduced death phase—the number of microbes decreases, so the death rate slows.
4. Molds.
5. Temperature, oxygen, relative humidity, water activity, pH, nutrients, and inhibitors.
6. They are harder to remove than normal dirt and microbes. They provide a place for microorganisms that cause spoilage and foodborne illness to grow. Chunks of biofilms can break off into food during processing.
7. Physical spoilage is obvious because of changes in the color, texture, smell, or flavor of the food. It may be aerobic (with oxygen) on the food surface or anaerobic (without oxygen) inside the food or in vacuum-packed or canned foods. Chemical spoilage is when enzymes in the food and from microorganisms break down protein, fats, carbohydrates, and other large molecules in the food into smaller and simpler compounds.
8. *Salmonella* spp., *Staphylococcus* spp., *Clostridium perfringens, Clostridium botulinum, Campylobacter* spp., *Listeria monocytogenes, Yersinia enterocolitica,* and *Escherichia coli* O157:H7.
9. Cereal grains, flour, bread, cornmeal, popcorn, peanuts, and peanut butter. Avoid insect damage and keep dry.
10. Destroying microbes means killing them so they cannot grow. Inhibiting microbes means injuring them or creating conditions in which they do not grow well; the microbes can recover and, if conditions become good, they can start to grow again.

CHAPTER 3

1. Infection—microorganism in food grows in the body after it is eaten.
   Intoxication—microorganism grows and produces toxins in the food before it is eaten.
2. Good source of protein, amino acids, B vitamins and minerals.
   Handled extensively.
   Frequently stored for long periods without refrigeration.
3. Food production and foodservice employees.
4. Disposable gloves, frequent handwashing.
5. Air filters, appropriate packaging.
6. Steam table, ice tray, transparent shield over and in front of food.
7. Covered, in clean containers, and away from food storage areas.

CHAPTER 4

1. Bathe daily, scrub fingernails daily, wash hair twice a week, wear clean clothes. Get enough sleep, eat well, and avoid exposure to infectious diseases when possible to cut down on illness.
2. At the beginning of their shift; after using the toilet; after handling garbage or anything dirty; after handling raw meats, poultry, eggs, or dairy products; after handling money; after smoking; and after coughing and sneezing.
3. Everyone carries bacteria that can cause foodborne illness, even when they are healthy. Employees can carry diseases even if they do not have symptoms or have recovered.
4. Skin, hands, fingernails, hair, mouth, nose, waste organs.
5. Jewelry is a safety hazard if it is caught in machinery. Jewelry may be contaminated and may fall into or come in contact with food.
6. Set a good example, educate and train employees, check employees health and skin condition, inspect employees work habits, provide good toilet and handwashing facilities, provide disposable gloves, provide good laundry facilities.

CHAPTER 5

1. Smooth, hard, nonporous.
2. Step 1: Separating soil from the surface or equipment.
   Step 2: Dispersing soil in the cleaning solution.
   Step 3: Preventing dispersed soil from reattaching to the surface.
3. Breaks up soil into smaller particles or droplets that can be dispersed and carried away by the cleaning solution; stops dispersed soil from settling.
4. Food particles protect microbes from sanitizers.
5. Free of minerals (i.e., soft water), free of microorganisms, clear, colorless, noncorrosive.
6. Alkaline cleaners remove organic soils, e.g., fat, oils, grease, protein. Acid cleaners remove inorganic soils, e.g., minerals, encrusted mineral scale.

7. Away from traffic and food supplies; off the floor; in a cool, dry, locked, fire-safe cupboard; acid and alkaline cleaners separated; in clearly labeled, sealed containers.
8. Flush the skin or eyes with plenty of water for 15 to 20 mintues.

CHAPTER 6

1. Make sure they are thoroughly clean.
2. Killing or removing all microbes.
3. Heat, radiation, chemicals. Chemicals are used most often in food-processing facilities.
4. Expensive, uses a lot of energy, inefficient, easy to mistake water vapor for steam, can make soil cake onto surface, need to check that the temperature stays hot enough.
5. Fruits, vegetables, and spices.
6. The temperature of the solution must be less than 55°C (131°F), and the soil must be light.
7. Destroys all types of vegetative bacteria, yeasts, and molds quickly.

   Works well in different environments (i.e., soiled surfaces, hard water, different pHs, soap or detergent residues).

   Cleans well.

   Dissolves in water.

   Stable as purchased (concentrate) and as used (diluted).

   Easy to use.

   Readily available.

   Inexpensive.

   Easy to measure.

   Does not irritate skin, nontoxic.

   No offensive odor.
8. Chlorine: Works on a broad range of microorganisms, cheap, no rinsing required at 200 ppm or less, available as liquid or granules, not affected by hard water.

   Iodophors: Stable at low pH, most effective sanitizer for viruses, can be used at very low concentrations, not affected by hard water, stain unremoved organic and mineral soils, do not irritate skin.

   Quats: Penetrate well, natural wetting agents, Good for *L. monocytogenes* and mold, stable in presence of organic soil, not corrosive or irritating, heat stable.

   Acid: Neutralizes excess alkalinity from cleaning, can combine with final rinse, does not corrode metals, works against yeasts, stable when heated and in presence of organic matter.

CHAPTER 7

1. Labor.
2. Mechanized cleaning systems, reuse water and cleaning solutions, lower temperatures, use the right amount of chemical at the right concentration, use the right chemical for the type of surface or equipment to reduce corrosion, good overall management.

3. Fragments can get stuck and can corrode surfaces or contaminate food.
4. Fog makes it hard to see, condensation can encourage mold growth, steam can burn workers.
5. The force helps clean the equipment or surface and gets the cleaning solution into hard-to-reach areas.
6.

Centralized	*Portable*
Advantages	
Outlets are available at several places in the facility. Reduces labor costs.	Workers can move the equipment to any area that needs to be cleaned. Equipment is fairly cheap to buy and maintain. Works well against *Listeria monocytogenes.*
Disadvantages	
Equipment is expensive to buy and maintain. System must be custom designed.	Takes more labor to move equipment to the area where it is needed. Less automated than centralized units. Less durable than centralized units.

7. Foam clings to the surface, so the cleaner does not need to be reapplied several times, and the worker can see where the foam is and is unlikely to clean the same area twice.
8. CIP equipment is not taken apart; built-in heads spray cleaning and rinsing solutions into the equipment, and built-in drains remove and/or recirculate the solutions; works well for fluid processing (e.g., dairy and beverages), does not work well for heavy soil; system needs to be custom designed and is expensive to buy and maintain; a microprocessor unit can control the operation. COP equipment is taken apart and cleaned in another area; works well for heavily soiled equipment; less expensive than CIP systems to buy and maintain; works well for small parts and utensils.

CHAPTER 8

1. It attracts insects and rodents, smells unpleasant, becomes a public nuisance, and looks unattractive.
2. Microbes grow on the food and use up the oxygen dissolved in the water. When the amount of dissolved oxygen in the water falls below 5 ppm, the fish die.
3. BOD (biochemical oxygen demand) is the amount of oxygen waste uses as it decomposes.
4. Checking water balance, sampling wastewater, and checking the amount of pollution.

5. BOD (biological oxygen demand), COD (chemical oxygen demand), DO (dissolved oxygen), TOC (total organic carbon), SS (settleable solids), TSS (total suspended solids), TDS (total dissolved solids), and FOG (fats, oils, and grease).
6. Truck to municipal garbage dump, compost, dry and grind for animal feed.
7. Advantages: Grease and solid materials recovered from waste can be sold, may reduce municipal surcharges; fewer complaints from the municipal treatment facility.

   Disadvantages: Pretreatment facilities are expensive and make the processing operation more complex; wastewater treatment can be expensive to maintain, monitor, and document; pretreatment facilities may be subject to property tax.
8. Pretreatment reduces the levels of pollutants in the wastewater to meet municipal regulations. Primary treatment removes particles from the wastewater. Secondary treatment oxidizes dissolved organic matter. Tertiary treatment removes pollutants such as colors, smells, brines, and flavors. Disinfection removes most of the remaining microbes from the water.

CHAPTER 9

1. Cockroaches, houseflies, and fruit flies.
2. Pests carry microorganisms in and on their bodies. They transfer them to food, surfaces, and equipment when they walk on them, bite or chew them, spit saliva onto them, or leave droppings on them.
3. Walk into a dark room and turn on the lights; check for a strong, oily odor; or look for their small, black or brown, spherical feces.
4. Good sanitation, fill cracks, inspect deliveries for adult cockroaches and eggs.
5. Keep flies out using air screens, mesh screens, double doors, and self-closing doors. Remove garbage promptly, and keep it covered and enclosed so that it does not attract flies. Use electric traps.
6. Residual pesticides work for a while after they are applied. Nonresidual pesticides only work for a short time when they are used.
7. Prevent them from entering; make sure there are no places for them to live; remove food sources; kill them by gassing, trapping, poisoning; tracking powder.
8. • Make sure all pesticide containers have clear labels.
   • Check that exterminators have work insurance.
   • Follow instructions when using pesticides.
   • Use the weakest poison that will destroy the pests.
   • Use oil-based and water-based sprays in appropriate places.
   • Wear protective clothing while applying pesticides, and wash hands afterwards.
   • Be careful not to contaminate food, equipment, and utensils with pesticides.
   • Call a doctor or poison control center if accidental poisoning occurs.
9. Use of a combination of several methods to control pests effectively and economically with minimal use of pesticides and minimal effects on the environment. IPM includes inspection, housekeeping, mechanical and physical methods, and chemical and biological methods.

CHAPTER 10

1. To make sure that the firm produces safe and acceptable food while maintaining productivity and efficiency.
2. a. Clear objectives and policies.
   b. Sanitation requirements for processes and products.
   c. An inspection system that includes procedures.
   d. Specified microbial, physical, and chemical values for products.
   e. Procedures and requirements for microbial, physical, and chemical testing.
   f. A personnel structure, including an organizational chart for the QA program.
   g. A QA budget to cover related expenses.
   h. A job description for all QA positions.
   i. An appropriate salary structure to attract and retain qualified QA staff.
   j. Constant supervision of the QA program and regular reports of the results.
3. Inspect facilities and equipment, develop specifications and standards, sample and test products, check staff hygiene, meet government regulations, evaluate cleaning programs, report results, train staff, troubleshoot and correct problems, keep accurate records.
4. Exact, e.g., temperature, pH. Subjective: e.g., flavor, smell.
5. A critical control point is any point in food production where microorganisms should be destroyed or controlled to prevent a hazard. A control point is any point during food production where loss of control may result in an economic or quality defect, or the low probability of a health risk.
6. Because microbial tests often slow, and results may not be available for hours or even days. Microbial tests may be used to make sure the HACCP program is working properly, i.e., verification.

CHAPTER 11

1. *Listeria monocytogenes.* Survives aerobic and anaerobic conditions, grows at refrigeration temperatures, found in animal intestines and some plants; contamination can come from several sources, can survive pasteurization if contamination is heavy.
2. Water supply, waste disposal and drainage facilities, energy supply, proximity to highways, proximity to milk supply, proximity to a large populated area.
3. Minerals, lipids, carbohydrates, proteins, and water. Dust, lubricants, microorganisms, cleaning compounds, and sanitizers.
4. Presoften the water or use additives to the cleaning compound to overcome the hardness.
5. Save labor and energy; only a small amount is needed.
6. Heat products for the minimum amount of time at the minimum temperature possible, cool product heating surfaces before and after emptying processing vats, and keep soil films moist by rinsing away foam and other debris and leaving water in the processing vats until cleaning.

CHAPTER 12

1. *L. monocytogenes* can grow at refrigerator temperatures.
2. • Scrub floors and drains every day. Rinse drains with disinfectant every day.
   • Scrub walls each week.
   • Clean the outside of all equipment, light fixtures, sills and ledges, piping, vents, and other processing and packaging areas that are not part of the daily cleaning program.
   • Clean cooling and heating units and ducts each week.
   • Caulk all cracks in walls, ceilings, and window and door sills.
   • Reduce condensate on ceilings.
   • Scrub and clean raw-material areas as often as the processing and packaging areas.
   • Keep hallways and passageways clean and dry.
   • Minimize traffic in and out of processing and packaging areas. Establish plant traffic patterns to reduce cross-contamination from feet, containers, pallet jacks, pallets, and fork trucks.
   • Change outer clothing and sanitize hands, or change gloves when moving from an area containing raw ingredients to an area containing finished product.

     Change into clean work clothes each day at the plant. Color-code clothing to show which clothes belong in each area.
   • Make sure visitors change into clean clothes provided at the plant.
   • Monitor the surfaces and air to make sure that *Listeria* are under control.
   • Enclose processing and packaging rooms so that they only receive filtered air.
   • Clean and sanitize all equipment and containers before bringing them into processing and packaging areas.
3. • Microorganisms change the color and flavor of meat and poultry products.
   • Longer shelf life for self-service merchandise.
   • Less meat and poultry is wasted because it is discolored or spoiled.
   • Sanitary conditions improve the image and reputation of a firm.
   • Regulatory agencies and consumers insist on good sanitation.
   • Sanitary and tidy surroundings improve morale and productivity of employees.
   • More processing and handling of food require a stricter sanitation program.
   • Sanitation is good business.
4. Pick up debris, prerinse and wet, wash with cleaning agent, rinse, inspect, sanitize, and prevent recontamination.
5. CCP1—skinning: The hide is very contaminated, and there is no good way to remove soil from the live animal before slaughter. Therefore, it is important to skin the animal to cut down on cross-contamination from the hide to the carcass.

   CCP2—prewash: This step should remove most of the contamination from contact with the hide and minimize attachment of microbes to the carcass.

   CCP3—bacterial wash: Bactericidal compounds (e.g., acetic acid) reduce contamination with microbes, including pathogens from the animal's intestines.

CCP4—evisceration: Animals hold large numbers of pathogens in their intestines. The best way to make sure these pathogens do not contaminate the carcass is to teach operators how to remove the intestines and internal organs without bursting them.

CCP5—final wash: This bactericidal rinse reduces the number of microbes on the carcass and minimizes the number of pathogens that are carried into the rest of the processing and packaging process.

CCP6—chill: Rapid chilling controls the growth of pathogens.

CCP7—storage: Strict temperature control and daily cleaning and sanitizing of equipment prevent pathogens from growing. The population of pathogens at this point shows how well the previous CCPs have controlled contamination. Maintaining the product temperature below 7°C (45°F) is a CCP to prevent growth of pathogens during packaging and distribution.

6. Changes in supervisor, training new employees, complicated procedures may be hard to remember.

CHAPTER 13

1. Chilling begins immediately after harvesting. Chilling reduces the temperature of the product to 10°C (18°F) within 4 hours. Chilling continues to approximately 1°C (34°F).
2. a. Cover electrical equipment with polyethylene sheeting.
   b. Remove large debris and put it in trash containers.
   c. Manually or mechanically remove soil from the walls and floors by scraping, brushing, or hosing with mechanized cleaning equipment. Start at the top of equipment and walls, and work down toward the floor drains or exit.
   d. Take the equipment apart.
   e. Prerinse with water at 40°C (104°F), or lower to wet the surfaces and remove large and water-soluble debris.
   f. Apply a cleaning compound that works on organic soil (usually an alkaline cleaner) using portable or centralized high-pressure, low-volume or foam equipment. The cleaning solution should not be hotter than 55°C (131°F).
   g. Leave the cleaning compound for about 15 minutes to work on the soil, and rinse the equipment and area with water at 55 to 60°C (131-140°F).
   h. Inspect the equipment and the facility, and reclean if necessary.
   i. Make sure that the plant is microbially clean by using a sanitizer.
   j. Avoid contaminating the equipment and the area during maintenance and setup.
3. • Good public-image first impressions for inspectors and the public.
   • The condition of the outside of the plant often reflects the plant's hygiene standards inside.
   • Wet areas may contaminate food products through seepage and by providing a place for microorganisms and insects to grow.
   • Very dusty roads, yards, or parking lots can contaminate areas where food is exposed.
   • Untidy refuse, litter, equipment, and uncut weeds or grass around the plant buildings may be a good place for rodents, insects, and other pests to breed.

4. • Use of uncontaminated wastewater from one area of a food-processing operation (e.g., water from the final rinse in a cleaning cycle) in other areas that do not need drinkable water
   • Use of closed water systems in food-processing operations in which all water used in processing is continuously filtered to remove solids
   • Use of dry conveyors to transport solids, instead of water
5. Pathogens in seafood are rare and testing methods are not very accurate or precise. HACCP methods are less expensive. HACCP identifies and controls factors that could cause a hazard, rather than just checking to see if the end product is safe.

CHAPTER 14

1. Heat.
2. Soil, air, pests.
3. Easier to keep clean, fewer employee injuries.
4. So that pests and microbes do not have places to live or enter the building.
5. High-volume plants, juice processors.
6. Traditional: Cans are filled, sealed, and then heated to sterilize. Aseptic: Food and containers are sterilized separately, food is cooled, then containers are filled and sealed in aseptic conditions.
7. So that he or she knows which tests to use.

CHAPTER 15

1. They are mostly sugars and dissolve in water.
2. There is no good way to detoxify a beverage after it is made.
3. a. Rinse to remove large particles and soil that is not stuck on, to wet the area, and to help the cleaning compound work properly.
   b. Apply a cleaning compound (usually as foam) to wet and penetrate the soil.
   c. Rinse to remove the dispersed soil and the cleaning compound.
   d. Sanitize to destroy microorganisms.
   e. Rinse again if using a quaternary ammonium sanitizer before the cleaned area touches any beverage products.
4. Flocculation, filtration (through a sandbed), chlorination, sterile filtration, reverse osmosis, activated carbon, and deionization
5. Advantages: Penetrates well, no chemical residues. Disadvantages: Expensive, can leave condensation, slower than chemicals.
6. Bottling.
7. Acidity, carbon dioxide content, sugar level, alcohol level.

CHAPTER 16

1. Foods that contain very little water, such as bakery goods, nuts and seeds, candy, cereal, grains, and snack foods.
2. Insects, rodents, mold.
3. Dust provides food for rodents, birds, insects, and microbes and can get into finished food products.
4. Control humidity and temperature, have good ventilation, inspect and clean raw ingredients and storage and processing areas regularly, make sure the building is well maintained so that pests cannot get in, keep the inside and outside of the building tidy.
5. Advantages: Kills pests; can be used to treat a piece of infested equipment.
   Disadvantages: Uses a lot of energy, expensive.
6. Hand brooms, push brooms, dust and wet mops, dustpans, vacuum cleaners (portable or installed), mechanical scrubber or sweeper, compressed air, cylindrical brushes.

CHAPTER 17

1. Wash food, protect food, keep food hot or cold.
2. Give them tasks where they do not touch food, or do not let them work until they are healthy.
3. • Cooling tables to keep foods at 2°C (36°F) or cooler.
   • Warming tables to keep foods at 60°C (140°F) or hotter.
   • Authorized workers to serve food in hygienic conditions; customers not to handle serving utensils.
   • Servers to wear hairnets or hats, and not to touch hair while serving.
   • Transparent shields between customers and service tables and dishes.
4. Multiple preparation steps, temperature changes, large volume, and contaminated basic foods.
5. The sanitizer can be neutralized by the dirt during cleaning, so it needs to be used after cleaning for it to be able to sanitize.
6. Air-dry, because wiping cloths can recontaminate them.
7. Employees need reminders to keep sanitation a top priority; employee turnover is high in the foodservice industry.

CHAPTER 18

1. Buying, receiving, storing, preparing, cooking, holding, serving, cooling, reheating. Cooling causes foodborne illness most often.
2. Refrigerated below 4°C (40°F); frozen below –18°C (0°F).
3. Within 6 hours.

4. 74°C (165°F), 15 seconds.
5. In refrigerator, cooking from frozen, in microwave.
6. Close to work stations, foot-operated faucets, powder or liquid soap, and individual towels or air dryers.

CHAPTER 19

1. Managers, quality assurance (QA), research and development, plant engineers, purchasing department, sanitation employees.
2. Delegate, train, supervise.
3. Training, professional treatment, job enrichment program, total quality management (with employee input); use good hygiene themselves.
4. To help employees clean efficiently and quickly; waste less time, energy, and cleaning materials; prevent injuries; use less water; and produce less sewage.
5. To "sell" sanitation to managers and employees, to help the company have a good public image.
6. TQM helps management and employees work together for better productivity, lower costs, and more consistent and more hygienic products.